

COUPE DU MONDE FIFA 2026

Prevision complete — moteur PELE (Nate Silver)

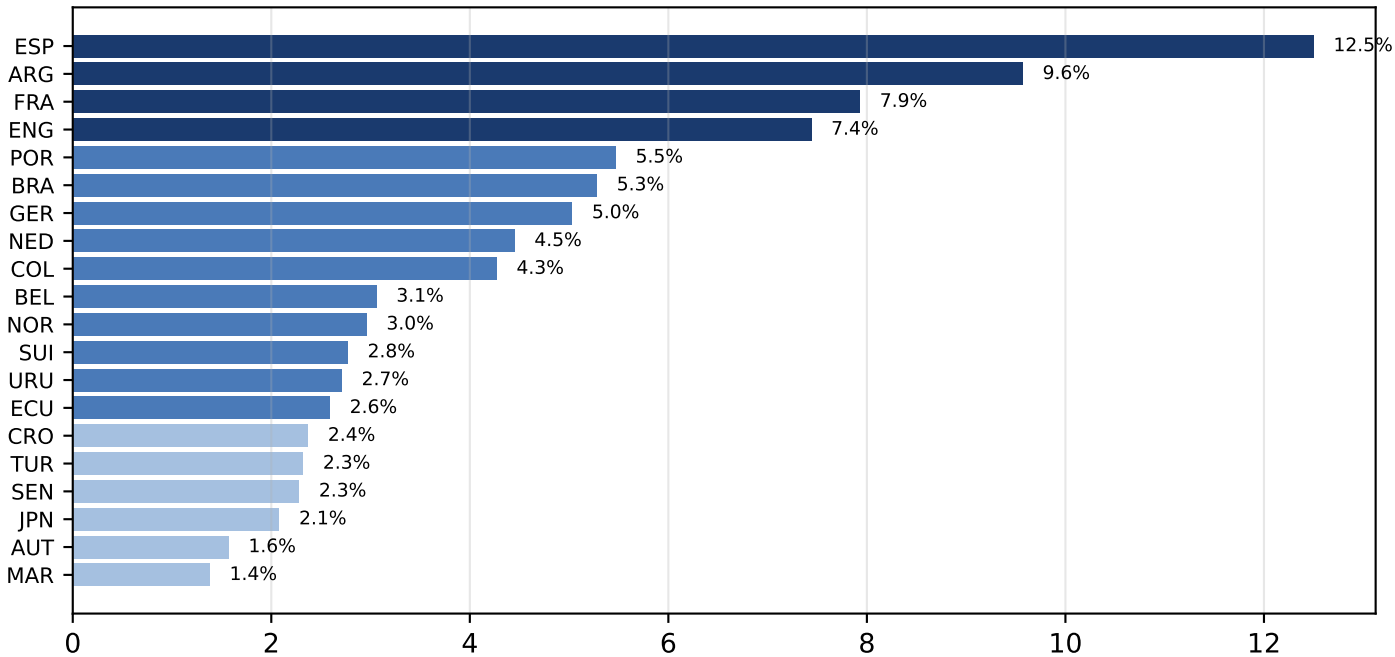
Etats-Unis · Canada · Mexique | 48 nations · 12 poules · 104 matchs

Monte Carlo : 10,000 simulations completes (poules + KO)

TOP 10 CONTENDERS — probabilite de gagner le tournoi

#	Nation	Champion	Finale	1/2	1/4	1/8
1	ESP	12.5%	19.4%	29.5%	43.6%	64.3%
2	ARG	9.6%	16.0%	25.6%	40.1%	58.1%
3	FRA	7.9%	13.6%	23.1%	37.1%	57.8%
4	ENG	7.4%	13.2%	22.3%	36.7%	58.1%
5	POR	5.5%	10.4%	17.9%	31.7%	52.7%
6	BRA	5.3%	9.9%	18.6%	32.2%	52.8%
7	GER	5.0%	9.5%	18.0%	31.5%	56.3%
8	NED	4.5%	8.3%	16.5%	29.6%	49.3%
9	COL	4.3%	8.5%	15.7%	27.7%	49.1%
10	BEL	3.1%	6.6%	12.8%	26.6%	49.5%

Distribution complete P(Champion) — top 20



P(Champion) en %

Genere par scripts/pele_v5_forecast.py — phase test, hors prod

Methodologie

Moteur, donnees, calibration et limitations connues

1. Moteur de base

PELE de Nate Silver (natesilver.net/p/pele-methodology). Elo on h-margin Phase 1 + mean-reversion

Transfermarkt Phase 2.

2. Vraies lambdas Silver

Pour les 72 matchs phase poule, on utilise les lambdas officielles Silver (CSV paypal). 100% des matchs CDM matches.

3. WC shrink 0.9x

Methodo Silver : applique en group stage car les matchs CDM sont plus upsets que la default. Empiriquement valide : MAE vs Pinnacle 6.23 -> 5.27.

4. KO : formule calibree

baseline=1.35, scale=1.2, alpha_tilt=0.2 (RMSE 0.244 buts vs vraies lambdas Silver sur 72 matchs). Tilt ignore en KO (biais mineur).

5. Tiebreakers FIFA

Implementation officielle (FIFA WC 2026 art.13) : H2H pts/diff/buts recursif, fallback diff globale/buts globaux. Cards remplacees par Elo (Monte Carlo).

6. Meilleurs 3emes

8 sur 12 selon regles FIFA + bracket THIRD_PLACE_SLOTS officiel.

7. Bracket KO

R32 -> R16 -> QF -> SF -> Finale, slots officiels FIFA. Tirs au but : sigmoid sur Elo clamped [0.15, 0.85].

8. Monte Carlo

N = 10,000 sims completes. Seed = 42 (reproductible).

9. Comparatif V8 prod

On lance aussi N sims avec la formule V8 prod (baseline 1.25, scale 0.5) pour quantifier les divergences. Voir page 'Divergences V8 vs PELE'.

10. Limitations

Pas de Phase 2 dans notre patch (Transfermarkt mean-reversion absente du KO). Pas de Tilt en KO. Roster

Transfermarkt date snapshot 2025-Q2. Bookmakers Pinnacle : 19/104 matchs couverts (phase poule seulement).

Poule A

MEX · RSA · KOR · CZE

Composition + Sim phase poule (10k MC)

Eq.	PELE	Tilt	V8 prod	P-V8	pts moy	P(1eP)	P(qualif R3)
MEX	1824	-0.230	1857	-34	4.62	36.8%	78.7%
RSA	1665	-0.220	1548	+117	3.41	14.1%	52.4%
KOR	1755	-0.150	1726	+29	4.07	21.7%	67.9%
CZE	1790	+0.130	1703	+87	4.35	27.4%	72.6%

Les 6 matchs de la poule (vraies probas Silver + WC 0.9x)

Match	λ H	λ A	P(H)	P(N)	P(A)	modal	Pin H/N/A
MEX - RSA	2.02	0.48	0%	0%	0%	2-0	62/23/16
KOR - CZE	1.21	1.29	33%	30%	37%	1-1	36/30/34
MEX - KOR	1.83	0.57	0%	0%	0%	2-1	—
CZE - RSA	1.56	0.93	0%	0%	0%	2-1	—
CZE - MEX	0.67	1.93	10%	24%	66%	1-2	—
RSA - KOR	0.92	1.28	24%	31%	44%	1-1	—

Progression KO sur 10k simulations (en %)

Eq.	1/16	1/8	1/4	1/2	Finale	Champion
MEX	78.7%	40.3%	17.5%	7.7%	3.0%	1.08%
RSA	52.4%	19.8%	6.3%	2.1%	0.7%	0.19%
KOR	67.9%	30.4%	11.5%	4.1%	1.4%	0.51%
CZE	72.6%	34.4%	14.2%	5.7%	2.1%	0.90%

Poule B

CAN · BIH · QAT · SUI

Composition + Sim phase poule (10k MC)

Eq.	PELE	Tilt	V8 prod	P-V8	pts moy	P(1e#)	(qualif R3)
CAN	1803	-0.070	1712	+91	4.61	27.9%	79.9%
BIH	1703	+0.210	1500	+203	3.85	16.7%	64.5%
QAT	1538	+0.240	1449	+88	2.72	7.8%	34.0%
SUI	1903	+0.140	1826	+77	5.35	47.7%	87.4%

Les 6 matchs de la poule (vraies probas Silver + WC 0.9x)

Match	λ H	λ A	P(H)	P(N)	P(A)	modal	Pin H/N/A
CAN - BIH	1.93	0.67	0%	0%	0%	2-1	—
QAT - SUI	0.54	3.06	3%	13%	84%	1-3	11/17/72
CAN - QAT	2.87	0.43	0%	0%	0%	3-0	—
SUI - BIH	2.13	0.87	0%	0%	0%	2-1	—
SUI - CAN	1.20	1.20	33%	31%	36%	1-1	—
BIH - QAT	2.22	0.87	0%	0%	0%	2-1	—

Progression KO sur 10k simulations (en %)

Eq.	1/16	1/8	1/4	1/2	Finale	Champion
CAN	79.9%	41.2%	17.9%	7.5%	3.0%	1.15%
BIH	64.5%	27.3%	9.9%	3.3%	1.1%	0.37%
QAT	34.0%	10.1%	2.5%	0.6%	0.1%	0.01%
SUI	87.4%	53.2%	27.0%	13.2%	6.2%	2.77%

Poule C

BRA · MAR · HAI · SCO

Composition + Sim phase poule (10k MC)

Eq.	PELE	Tilt	V8 prod	P-V8	pts moy	P(1eP)	(qualif R3)
BRA	1990	-0.070	2030	-40	5.39	48.0%	88.2%
MAR	1851	-0.440	1807	+44	4.33	24.5%	75.6%
HAI	1622	+0.230	1454	+168	2.71	8.2%	35.0%
SCO	1815	+0.040	1734	+81	4.11	19.3%	70.4%

Les 6 matchs de la poule (vraies probas Silver + WC 0.9x)

Match	λ H	λ A	P(H)	P(N)	P(A)	modal	Pin H/N/A
BRA - MAR	1.65	0.65	0%	0%	0%	2-1	59/25/16
HAI - SCO	0.87	2.04	15%	24%	60%	1-2	15/19/67
BRA - HAI	3.06	0.54	0%	0%	0%	3-1	—
SCO - MAR	0.92	1.19	29%	33%	38%	1-1	—
SCO - BRA	0.77	2.03	12%	23%	65%	1-2	—
MAR - HAI	1.83	0.67	0%	0%	0%	2-1	—

Progression KO sur 10k simulations (en %)

Eq.	1/16	1/8	1/4	1/2	Finale	Champion
BRA	88.2%	52.8%	32.2%	18.6%	9.9%	5.28%
MAR	75.6%	36.0%	17.9%	8.0%	3.5%	1.38%
HAI	35.0%	10.7%	3.5%	1.1%	0.3%	0.10%
SCO	70.4%	31.8%	14.9%	6.5%	2.6%	0.90%

Poule D

USA · PAR · AUS · TUR

Composition + Sim phase poule (10k MC)

Eq.	PELE	Tilt	V8 prod	P-V8	pts moy	P(1eP)	(qualif R3)
USA	1786	+0.100	1856	-70	3.87	21.8%	66.7%
PAR	1834	-0.230	1792	+42	4.20	23.9%	70.7%
AUS	1781	-0.140	1722	+59	3.79	18.1%	58.7%
TUR	1888	+0.280	1937	-50	4.60	36.1%	78.0%

Les 6 matchs de la poule (vraies probas Silver + WC 0.9x)

Match	λ H	λ A	P(H)	P(N)	P(A)	modal	Pin H/N/A
USA - PAR	1.28	1.01	0%	0%	0%	1-1	49/25/25
AUS - TUR	0.86	1.94	15%	25%	60%	1-2	20/24/55
USA - AUS	1.66	0.84	0%	0%	0%	2-1	—
TUR - PAR	1.58	1.12	0%	0%	0%	2-1	—
TUR - USA	1.40	1.31	37%	29%	34%	1-1	—
PAR - AUS	1.47	0.93	0%	0%	0%	1-1	—

Progression KO sur 10k simulations (en %)

Eq.	1/16	1/8	1/4	1/2	Finale	Champion
USA	66.7%	30.8%	13.0%	5.1%	1.9%	0.71%
PAR	70.7%	34.8%	15.5%	7.0%	3.0%	1.37%
AUS	58.7%	26.5%	11.2%	4.3%	1.5%	0.54%
TUR	78.0%	43.5%	21.1%	10.4%	4.8%	2.32%

Poule E

GER · CUW · CIV · ECU

Composition + Sim phase poule (10k MC)

Eq.	PELE	Tilt	V8 prod	P-V8	pts moy	P(1eP)	qualif R3
GER	1969	+0.600	1928	+41	5.41	47.0%	89.3%
CUW	1549	+0.110	1468	+81	2.41	5.8%	28.1%
CIV	1741	-0.080	1730	+11	3.77	15.5%	62.9%
ECU	1903	-0.240	1868	+35	4.97	31.7%	84.1%

Les 6 matchs de la poule (vraies probas Silver + WC 0.9x)

Match	λ H	λ A	P(H)	P(N)	P(A)	modal	Pin H/N/A
GER - CUW	3.63	0.57	0%	0%	0%	4-1	—
CIV - ECU	0.75	1.65	17%	28%	55%	1-2	27/34/38
GER - CIV	2.50	0.79	0%	0%	0%	3-1	—
ECU - CUW	2.49	0.51	0%	0%	0%	2-1	—
ECU - GER	1.23	1.77	25%	27%	48%	1-2	—
CUW - CIV	0.77	2.03	13%	24%	64%	1-2	—

Progression KO sur 10k simulations (en %)

Eq.	1/16	1/8	1/4	1/2	Finale	Champion
GER	89.3%	56.3%	31.5%	18.0%	9.5%	5.03%
CUW	28.1%	8.0%	1.8%	0.4%	0.1%	0.02%
CIV	62.9%	26.8%	10.1%	4.2%	1.5%	0.49%
ECU	84.1%	46.9%	23.4%	12.0%	5.5%	2.59%

Poule F

NED · JPN · SWE · TUN

Composition + Sim phase poule (10k MC)

Eq.	PELE	Tilt	V8 prod	P-V8	pts moy	P(1eR)	qualif R3
NED	1965	+0.280	1971	-6	5.08	43.2%	84.6%
JPN	1874	+0.100	1859	+15	4.43	27.2%	76.2%
SWE	1798	+0.220	1738	+60	3.88	18.4%	64.3%
TUN	1700	-0.250	1628	+72	3.11	11.2%	44.7%

Les 6 matchs de la poule (vraies probas Silver + WC 0.9x)

Match	λ H	λ A	P(H)	P(N)	P(A)	modal	Pin H/N/A
NED - JPN	1.77	1.14	0%	0%	0%	2-1	45/28/27
SWE - TUN	1.66	0.94	0%	0%	0%	2-1	49/28/23
NED - SWE	2.14	0.96	0%	0%	0%	2-1	—
TUN - JPN	0.77	1.94	13%	24%	63%	1-2	—
TUN - NED	0.59	2.30	8%	20%	73%	1-2	—
JPN - SWE	1.77	1.14	0%	0%	0%	2-1	—

Progression KO sur 10k simulations (en %)

Eq.	1/16	1/8	1/4	1/2	Finale	Champion
NED	84.6%	49.3%	29.6%	16.5%	8.3%	4.45%
JPN	76.2%	39.5%	20.4%	9.9%	4.5%	2.08%
SWE	64.3%	29.3%	13.5%	5.5%	2.1%	0.74%
TUN	44.7%	17.1%	6.4%	2.3%	0.7%	0.24%

Poule G

BEL · EGY · IRN · NZL

Composition + Sim phase poule (10k MC)

Eq.	PELE	Tilt	V8 prod	P-V8	pts moy	P(1eP)	(qualif R3)
BEL	1914	+0.270	1930	-16	5.31	48.2%	86.1%
EGY	1744	-0.190	1709	+36	3.99	20.6%	69.5%
IRN	1740	-0.180	1704	+36	3.99	19.5%	67.0%
NZL	1626	-0.050	1563	+63	3.21	11.7%	48.0%

Les 6 matchs de la poule (vraies probas Silver + WC 0.9x)

Match	λ H	λ A	P(H)	P(N)	P(A)	modal	Pin H/N/A
BEL - EGY	1.94	0.86	0%	0%	0%	2-1	58/23/19
IRN - NZL	1.56	0.93	0%	0%	0%	2-1	—
BEL - IRN	2.03	0.77	0%	0%	0%	2-1	—
NZL - EGY	0.85	1.75	16%	26%	57%	1-2	—
NZL - BEL	0.61	2.68	6%	16%	78%	1-3	—
EGY - IRN	1.29	1.11	0%	0%	0%	1-1	—

Progression KO sur 10k simulations (en %)

Eq.	1/16	1/8	1/4	1/2	Finale	Champion
BEL	86.1%	49.5%	26.6%	12.8%	6.6%	3.06%
EGY	69.5%	30.7%	11.7%	4.1%	1.5%	0.49%
IRN	67.0%	29.0%	10.8%	3.7%	1.5%	0.35%
NZL	48.0%	17.4%	4.9%	1.4%	0.3%	0.11%

Poule H

ESP · CPV · KSA · URU

Composition + Sim phase poule (10k MC)

Eq.	PELE	Tilt	V8 prod	P-V8	pts moy	P(1e#)	(qualif R3)
ESP	2097	+0.180	1500	+597	6.24	61.3%	94.7%
CPV	1600	-0.250	1573	+27	2.75	6.2%	37.1%
KSA	1614	-0.220	1604	+10	2.83	6.8%	37.5%
URU	1907	-0.160	1866	+41	4.86	25.7%	83.9%

Les 6 matchs de la poule (vraies probas Silver + WC 0.9x)

Match	λ H	λ A	P(H)	P(N)	P(A)	modal	Pin H/N/A
ESP - CPV	3.24	0.36	0%	0%	0%	3-0	—
KSA - URU	0.49	2.21	7%	21%	73%	0-2	17/24/58
ESP - KSA	3.33	0.37	0%	0%	0%	3-0	—
URU - CPV	2.11	0.49	0%	0%	0%	2-0	—
URU - ESP	0.76	1.84	15%	26%	60%	1-2	—
CPV - KSA	1.00	1.00	33%	33%	33%	1-1	—

Progression KO sur 10k simulations (en %)

Eq.	1/16	1/8	1/4	1/2	Finale	Champion
ESP	94.7%	64.3%	43.6%	29.5%	19.4%	12.50%
CPV	37.1%	10.4%	3.1%	0.8%	0.2%	0.07%
KSA	37.5%	10.9%	3.7%	1.0%	0.3%	0.08%
URU	83.9%	41.5%	23.3%	11.3%	5.8%	2.71%

Poule I

FRA · SEN · IRQ · NOR

Composition + Sim phase poule (10k MC)

Eq.	PELE	Tilt	V8 prod	P-V8	pts moy	P(1eP)	(qualif R3)
FRA	2036	-0.020	2119	-83	5.33	47.3%	87.9%
SEN	1881	-0.350	1852	+29	4.16	20.2%	72.5%
IRQ	1640	-0.220	1564	+76	2.54	6.8%	30.3%
NOR	1926	+0.250	1952	-26	4.51	25.7%	78.6%

Les 6 matchs de la poule (vraies probas Silver + WC 0.9x)

Match	λ H	λ A	P(H)	P(N)	P(A)	modal	Pin H/N/A
FRA - SEN	1.65	0.75	0%	0%	0%	2-1	66/20/13
IRQ - NOR	0.61	2.68	5%	17%	78%	1-3	12/18/70
FRA - IRQ	2.87	0.43	0%	0%	0%	3-0	—
NOR - SEN	1.48	1.02	0%	0%	0%	1-1	—
NOR - FRA	1.14	1.77	23%	27%	50%	1-2	—
SEN - IRQ	1.93	0.57	0%	0%	0%	2-1	—

Progression KO sur 10k simulations (en %)

Eq.	1/16	1/8	1/4	1/2	Finale	Champion
FRA	87.9%	57.8%	37.1%	23.1%	13.6%	7.93%
SEN	72.5%	39.7%	21.1%	10.2%	4.9%	2.28%
IRQ	30.3%	10.4%	3.3%	1.0%	0.2%	0.03%
NOR	78.6%	44.1%	24.2%	12.6%	6.3%	2.96%

Poule J

ARG · ALG · AUT · JOR

Composition + Sim phase poule (10k MC)

Eq.	PELE	Tilt	V8 prod	P-V8	pts moy	P(1e#)	qualif R3
ARG	2065	-0.180	2116	-51	5.80	55.8%	91.2%
ALG	1769	+0.160	1804	-35	3.66	14.6%	60.7%
AUT	1851	+0.270	1500	+351	4.24	21.4%	73.2%
JOR	1658	-0.030	1610	+48	2.88	8.1%	38.6%

Les 6 matchs de la poule (vraies probas Silver + WC 0.9x)

Match	λ H	λ A	P(H)	P(N)	P(A)	modal	Pin H/N/A
ARG - ALG	2.50	0.60	0%	0%	0%	2-1	66/22/12
AUT - JOR	2.13	0.87	0%	0%	0%	2-1	—
ARG - AUT	2.41	0.70	0%	0%	0%	2-1	—
JOR - ALG	0.95	1.76	18%	27%	55%	1-2	—
JOR - ARG	0.44	3.06	2%	12%	86%	0-3	—
ALG - AUT	1.32	1.59	29%	28%	44%	1-2	—

Progression KO sur 10k simulations (en %)

Eq.	1/16	1/8	1/4	1/2	Finale	Champion
ARG	91.2%	58.1%	40.1%	25.6%	16.0%	9.57%
ALG	60.7%	24.3%	10.5%	4.0%	1.7%	0.53%
AUT	73.2%	32.5%	16.2%	7.6%	3.5%	1.57%
JOR	38.6%	12.4%	4.4%	1.4%	0.4%	0.18%

Poule K

POR · COD · UZB · COL

Composition + Sim phase poule (10k MC)

Eq.	PELE	Tilt	V8 prod	P-V8	pts moy	P(1eP)	(qualif R3)
POR	1986	+0.100	2060	-74	5.16	44.1%	85.8%
COD	1718	-0.110	1648	+70	3.25	11.5%	49.9%
UZB	1701	-0.050	1677	+24	3.13	10.3%	45.3%
COL	1952	-0.230	1974	-22	5.00	34.0%	84.0%

Les 6 matchs de la poule (vraies probas Silver + WC 0.9x)

Match	λ H	λ A	P(H)	P(N)	P(A)	modal	Pin H/N/A
POR - COD	2.21	0.69	0%	0%	0%	2-1	71/18/11
UZB - COL	0.59	2.21	7%	21%	72%	1-2	12/21/66
POR - UZB	2.40	0.60	0%	0%	0%	2-1	—
COL - COD	2.02	0.58	0%	0%	0%	2-1	—
COL - POR	1.20	1.20	33%	31%	36%	1-1	—
COD - UZB	1.39	1.11	0%	0%	0%	1-1	—

Progression KO sur 10k simulations (en %)

Eq.	1/16	1/8	1/4	1/2	Finale	Champion
POR	85.8%	52.7%	31.7%	17.9%	10.4%	5.47%
COD	49.9%	18.4%	6.7%	2.4%	0.9%	0.35%
UZB	45.3%	16.3%	5.6%	1.9%	0.7%	0.18%
COL	84.0%	49.1%	27.7%	15.7%	8.5%	4.27%

Poule L

ENG · CRO · GHA · PAN

Composition + Sim phase poule (10k MC)

Eq.	PELE	Tilt	V8 prod	P-V8	pts moy	P(1eR)	qualif R3
ENG	2031	-0.090	2015	+16	5.66	52.0%	90.2%
CRO	1909	+0.080	1925	-16	4.74	28.1%	80.7%
GHA	1641	-0.140	1632	+9	2.85	8.5%	38.9%
PAN	1723	+0.010	1628	+95	3.34	11.4%	53.3%

Les 6 matchs de la poule (vraies probas Silver + WC 0.9x)

Match	λ H	λ A	P(H)	P(N)	P(A)	modal	Pin H/N/A
ENG - CRO	1.84	0.76	0%	0%	0%	2-1	58/23/19
GHA - PAN	1.02	1.48	24%	29%	47%	1-1	46/25/29
ENG - GHA	2.77	0.43	0%	0%	0%	3-0	—
PAN - CRO	0.95	1.76	18%	27%	55%	1-2	—
PAN - ENG	0.60	2.40	6%	19%	75%	1-2	—
CRO - GHA	2.03	0.68	0%	0%	0%	2-1	—

Progression KO sur 10k simulations (en %)

Eq.	1/16	1/8	1/4	1/2	Finale	Champion
ENG	90.2%	58.1%	36.7%	22.3%	13.2%	7.44%
CRO	80.7%	43.2%	22.7%	11.5%	5.7%	2.37%
GHA	38.9%	12.2%	3.8%	1.1%	0.4%	0.07%
PAN	53.3%	20.3%	7.9%	3.0%	0.9%	0.21%

Meilleurs 3emes — distribution sur 10k sims

8 places parmi 12 poules selon FIFA art.13 (pts/diff/buts globale)

Poule	Eq.	P(3e)	P(qualif via 3e)	P(qualif 3e)
I	SEN	34.0%	26.5%	78%
C	SCO	33.0%	24.7%	75%
E	CIV	37.9%	24.3%	64%
B	BIH	34.7%	23.2%	67%
F	SWE	32.3%	22.2%	69%
G	IRN	30.6%	21.7%	71%
J	ALG	34.1%	21.7%	64%
D	PAR	26.4%	21.0%	79%
A	KOR	28.4%	20.9%	74%
L	PAN	36.1%	20.7%	57%
D	AUS	27.6%	19.7%	71%
A	RSA	27.6%	19.5%	71%
I	NOR	25.1%	19.1%	76%
C	MAR	25.9%	19.0%	73%
D	USA	25.9%	19.0%	73%
K	COD	34.5%	18.9%	55%
G	NZL	27.9%	18.2%	65%
G	EGY	25.8%	18.1%	70%
B	CAN	22.8%	17.8%	78%
K	UZB	29.4%	17.8%	61%
F	JPN	23.5%	17.7%	75%
F	TUN	26.9%	16.6%	62%
J	AUT	24.6%	16.5%	67%
E	ECU	21.0%	16.4%	78%
L	CRO	21.8%	16.0%	73%
A	CZE	23.4%	15.7%	67%
A	MEX	20.6%	15.6%	75%
H	URU	21.1%	15.6%	74%
L	GHA	28.6%	15.4%	54%
K	COL	19.3%	15.3%	79%

Bracket R32 — affiches probables

Pour chaque slot du bracket FIFA officiel, top 3 affiches les + frequentes

Eq. (top 16 P(R16))	P(R16)	Adversaires R16 les + frequents
ESP	64.3%	CRO(9%) · COL(8%) · ENG(8%)
ARG	58.1%	TUR(7%) · BEL(6%) · PAR(6%)
ENG	58.1%	ESP(8%) · MEX(7%) · CZE(4%)
FRA	57.8%	GER(10%) · ECU(6%) · BRA(5%)
GER	56.3%	FRA(10%) · BRA(6%) · NOR(5%)
SUI	53.2%	POR(8%) · COL(6%) · NED(4%)
BRA	52.8%	GER(6%) · NOR(6%) · ECU(6%)
POR	52.7%	SUI(8%) · ESP(7%) · CAN(4%)
BEL	49.5%	TUR(6%) · ARG(6%) · PAR(4%)
NED	49.3%	CAN(4%) · CZE(4%) · SUI(4%)
COL	49.1%	ESP(8%) · SUI(6%) · AUT(3%)
ECU	46.9%	FRA(6%) · BRA(6%) · MAR(3%)
NOR	44.1%	BRA(6%) · GER(5%) · NED(3%)
TUR	43.5%	ARG(7%) · BEL(6%) · URU(4%)
CRO	43.2%	ESP(9%) · MEX(3%) · ARG(3%)
URU	41.5%	TUR(4%) · USA(3%) · PAR(3%)

Note : un adversaire affiche a 30% signifie qu'en 30% des sims, cette equipe arrive en R16 contre cet adversaire.

Progression complete — top 24 P(QF)

Cumul des probabilités par tour atteint

Eq.	R32	R16	QF	SF	Final	Champ	Bronze
ESP	95%	64%	43.6%	29.5%	19.4%	12.50%	6.9%
ARG	91%	58%	40.1%	25.6%	16.0%	9.57%	6.3%
FRA	88%	58%	37.1%	23.1%	13.6%	7.93%	5.7%
ENG	90%	58%	36.7%	22.3%	13.2%	7.44%	5.4%
BRA	88%	53%	32.2%	18.6%	9.9%	5.28%	5.3%
POR	86%	53%	31.7%	17.9%	10.4%	5.47%	4.4%
GER	89%	56%	31.5%	18.0%	9.5%	5.03%	4.6%
NED	85%	49%	29.6%	16.5%	8.3%	4.45%	4.6%
COL	84%	49%	27.7%	15.7%	8.5%	4.27%	4.0%
SUI	87%	53%	27.0%	13.2%	6.2%	2.77%	3.5%
BEL	86%	50%	26.6%	12.8%	6.6%	3.06%	3.1%
NOR	79%	44%	24.2%	12.6%	6.3%	2.96%	3.1%
ECU	84%	47%	23.4%	12.0%	5.5%	2.59%	3.4%
URU	84%	42%	23.3%	11.3%	5.8%	2.71%	2.8%
CRO	81%	43%	22.7%	11.5%	5.7%	2.37%	2.9%
TUR	78%	43%	21.1%	10.4%	4.8%	2.32%	2.7%
SEN	73%	40%	21.1%	10.2%	4.9%	2.28%	2.5%
JPN	76%	39%	20.4%	9.9%	4.5%	2.08%	2.8%
CAN	80%	41%	17.9%	7.5%	3.0%	1.15%	1.8%
MAR	76%	36%	17.9%	8.0%	3.5%	1.38%	1.9%
MEX	79%	40%	17.5%	7.7%	3.0%	1.08%	2.0%
AUT	73%	32%	16.2%	7.6%	3.5%	1.57%	1.9%
PAR	71%	35%	15.5%	7.0%	3.0%	1.37%	1.8%
SCO	70%	32%	14.9%	6.5%	2.6%	0.90%	1.7%

Top contenders — detail 1-4

Pour les 4 favoris : ratings, top adversaires KO, chemin probable

#20 ESP PELE 2097 Tilt +0.180 V8 1500 Δ +597

P(Champion) 12.50% P(Final) 19.4% P(SF) 29.5% P(QF) 43.6% P(R16) 64.3% P(R32) 94.7%

Quarts probables : vs BEL 5% vs TUR 3% vs SUI 2%

1/8 finale probables : vs CRO 9% vs COL 8% vs ENG 8%

#20 ARG PELE 2065 Tilt -0.180 V8 2116 Δ -51

P(Champion) 9.57% P(Final) 16.0% P(SF) 25.6% P(QF) 40.1% P(R16) 58.1% P(R32) 91.2%

Quarts probables : vs POR 5% vs SUI 4% vs COL 4%

1/8 finale probables : vs TUR 7% vs BEL 6% vs PAR 6%

#20 FRA PELE 2036 Tilt -0.020 V8 2119 Δ -83

P(Champion) 7.93% P(Final) 13.6% P(SF) 23.1% P(QF) 37.1% P(R16) 57.8% P(R32) 87.9%

Quarts probables : vs NED 4% vs BRA 3% vs ENG 3%

1/8 finale probables : vs GER 10% vs ECU 6% vs BRA 5%

#20 ENG PELE 2031 Tilt -0.090 V8 2015 Δ +16

P(Champion) 7.44% P(Final) 13.2% P(SF) 22.3% P(QF) 36.7% P(R16) 58.1% P(R32) 90.2%

Quarts probables : vs BRA 4% vs FRA 3% vs NOR 3%

1/8 finale probables : vs ESP 8% vs MEX 7% vs CZE 4%

Top contenders — detail 5-8

Pour les 4 favoris : ratings, top adversaires KO, chemin probable

#24 POR PELE 1986 Tilt +0.100 V8 2060 Δ -74

P(Champion) 5.47% P(Final) 10.4% P(SF) 17.9% P(QF) 31.7% P(R16) 52.7% P(R32) 85.8%

Quarts probables : vs ARG 5% vs BEL 3% vs TUR 2%

1/8 finale probables : vs SUI 8% vs ESP 7% vs CAN 4%

#24 BRA PELE 1990 Tilt -0.070 V8 2030 Δ -40

P(Champion) 5.28% P(Final) 9.9% P(SF) 18.6% P(QF) 32.2% P(R16) 52.8% P(R32) 88.2%

Quarts probables : vs ENG 4% vs FRA 3% vs GER 3%

1/8 finale probables : vs GER 6% vs NOR 6% vs ECU 6%

#24 GER PELE 1969 Tilt +0.600 V8 1928 Δ +41

P(Champion) 5.03% P(Final) 9.5% P(SF) 18.0% P(QF) 31.5% P(R16) 56.3% P(R32) 89.3%

Quarts probables : vs NED 4% vs BRA 3% vs ENG 2%

1/8 finale probables : vs FRA 10% vs BRA 6% vs NOR 5%

#24 NED PELE 1965 Tilt +0.280 V8 1971 Δ -6

P(Champion) 4.45% P(Final) 8.3% P(SF) 16.5% P(QF) 29.6% P(R16) 49.3% P(R32) 84.6%

Quarts probables : vs GER 4% vs FRA 4% vs ENG 2%

1/8 finale probables : vs CAN 4% vs CZE 4% vs SUI 4%

Divergences V8 prod vs PELE — P(qualif R32)

Top 20 ecarts absolus | positif = PELE > V8 | negatif = V8 > PELE

Equipe	P(R32) V8 prod	P(R32) PELE	Delta (pts)
ESP	52.3%	94.7%	▲ +42.4
AUT	35.0%	73.2%	▲ +38.2
KSA	65.6%	37.5%	▼ -28.0
CPV	61.6%	37.1%	▼ -24.5
ALG	77.5%	60.7%	▼ -16.8
JOR	51.8%	38.6%	▼ -13.2
BIH	55.1%	64.5%	▲ +9.4
QAT	43.0%	34.0%	▼ -9.0
PAN	44.6%	53.3%	▲ +8.7
COD	41.9%	49.9%	▲ +8.0
GHA	46.4%	38.9%	▼ -7.5
CIV	69.7%	62.9%	▼ -6.8
USA	72.8%	66.7%	▼ -6.1
PAR	64.9%	70.7%	▲ +5.8
HAI	29.5%	35.0%	▲ +5.4
AUS	53.9%	58.7%	▲ +4.8
RSA	48.2%	52.4%	▲ +4.3
KOR	72.2%	67.9%	▼ -4.2
CZE	68.6%	72.6%	▲ +4.0
MEX	82.6%	78.7%	▼ -3.9

Et sur P(Champion) — top 10 ecarts :

ESP	V8	0.08%	PELE	12.50%	delta	+12.42
FRA	V8	12.79%	PELE	7.93%	delta	-4.86
ARG	V8	13.54%	PELE	9.57%	delta	-3.97
POR	V8	8.10%	PELE	5.47%	delta	-2.63
BRA	V8	7.09%	PELE	5.28%	delta	-1.81
GER	V8	3.48%	PELE	5.03%	delta	+1.55
AUT	V8	0.04%	PELE	1.57%	delta	+1.53
USA	V8	2.11%	PELE	0.71%	delta	-1.40
MEX	V8	2.31%	PELE	1.08%	delta	-1.23
CRO	V8	3.44%	PELE	2.37%	delta	-1.07

Value bets vs Pinnacle — phase poule

19 matchs avec cotes Pinnacle (de-margees Buchdahl)

Match	PELE+0.9x H/N/A	Pinnacle H/N/A	Side	Edge (pts)
ARG-ALG	77/18/6	66/22/12	H	+10.6
MEX-RSA	72/22/6	62/23/16	H	+10.3
FRA-SEN	56/28/16	66/20/13	N	+7.7
USA-PAR	42/31/27	49/25/25	N	+5.8
NED-JPN	49/27/24	45/28/27	H	+3.2
POR-COD	70/21/9	71/18/11	N	+3.0
BRA-MAR	57/28/15	59/25/16	N	+2.9
ENG-CRO	60/26/15	58/23/19	N	+2.3
SWE-TUN	51/28/21	49/28/23	H	+1.9
BEL-EGY	59/25/16	58/23/19	N	+1.9

Edge > 0 = PELE estime cette issue plus probable que Pinnacle. Pas une recommandation de pari : PELE est ~5pts MAE moins précis que Pinnacle en moyenne, edge doit etre > 5pts pour etre serieux.

Limitations & sources

A lire avant toute interpretation

- **Modele PELE Phase 2 absente du KO**

Notre formule calibree (1.35/1.2) reproduit la transformation rating→ λ a ± 0.24 buts mais sans le mean-reversion Transfermarkt de la Phase 2. Pour la phase poule c'est ok (vraies λ Silver injectees), pour le KO ca peut sous-estimer l'effet roster.

- **Tilt rating ignore en KO**

wc_simulator prod ne connait que l'Elo. Le Tilt Silver (propension offensive/defensive) n'est applique qu'en phase poule via vraies λ . En KO, c'est neglige.

- **PELE rating snapshot**

Les ratings PELE chargees datent du dernier scrape datawrapper. Silver met a jour $\sim 1x/jour$, les ratings peuvent avoir bouge depuis.

- **Tirs au but**

Modelises par sigmoid Elo clampee [0.15, 0.85]. Pas de modelisation du gardien ou de la fatigue. Hypothese 50/50 si nul a la fin du temps reglementaire.

- **Bookmakers**

Pinnacle couvre uniquement 19/104 matchs (les + populaires). Le 'value bet' n'a de sens que sur ces matchs.

- **V8 prod (comparatif)**

Lance avec elo_map = pin_calibrated snapshot 2026-05-20. N'utilise pas les overrides forced manuels (CDM 2026 final roster pas encore connu).

- **Pas de blessures, suspensions, forme**

Le modele est purement statique sur les ratings de base. ECT (event change tag) non integre.

- **Phase 2 Silver = score de roster Transfermarkt**

Pour reproduire ESP a 99.7% comme Silver, V8 prod aurait besoin d'integrer un score de roster (valeur 23 joueurs Transfermarkt). Voir piste P2 dans le plan d'evolution.

Sources & fichiers

- Vraies probas Silver : live/data/pele_paywall/silver_projections.csv (72 matchs)
- Ratings PELE/Tilt : live/data/pele_cache/{pele,tilt,rr}.csv (211 nations)
- V8 prod snapshot : artifacts/football-lab/lab/data/snapshots/initial_baseline_2026-05-20/
- Cotes Pinnacle : pinnacle_wc2026_odds.json (snapshot 2026-05-20)
- Bracket FIFA officiel : importe depuis artifacts/football-dashboard/wc_simulator.py
- Methodo PELE : .agents/memory/pele-methodology.md + .local/refs/pele_paywall/methodology_clean.txt